Show Recorder

Reference manual — v0.2.15

Show Recorder is a multitrack recorder for front of house: every input of the show to its own file, a stereo mix you can hand to the band that night, and the stems for whoever mixes it later. It's built to be left running for a whole show and trusted, and to do the after-show work (finding the songs, mixing, mastering, exporting) in the same window.

This manual describes the first version of the app, during the beta. The short version is the **Quick Reference**, beside this manual in the download.

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1. The window

Show Recorder is laid out as Console Control is, top to bottom:

- **The header**: the app, the session and the version on the left, with the console MIDI Bridge is on ("v0.2.15 · Wing", "· DiGiCo", "· SQ"; dLive until a console is known); **Lock** in the centre (see section 5); the dylanmaudio mark on the right.
- **The transport row**: the clock, large; **Go to start**, **Play**, **Stop**, **Record** and **Loop**; the **LTC** button (click it to turn Follow timecode in playback on or off; right-click it for **Settings** → **Sync**); then the status chips: timecode, **REC** during a take, the disk, the dropout counter (see section 6), the load, the interface and its rate, **MIDI** and **Companion** (each opens its Settings tab).
- **The status card**: what Show Recorder is doing (Stopped, Playing, Recording — Take 3, Locked) and what that means (the channels armed and the time left on the disk; why it can't record; how many problems **Before You Record** has). Under it, the last message, with **×** to clear it.
- **The toolbar**, in two rows. The first: the views (**Tracks**, **Patch Matrix**, **Mixer**, **Processing**), **Undo** and **Redo**, and **Snap**. The second: **Sync from Console**, **Before You Record**, **Settings**, then **Marks** (**+ Marker**, **+ Region**, **List**) and **the side panel button** (the side panel).

- **The timeline**, with the track list on the left. Its header has **Arm** (all), **+ Track** (an input channel, a mix from the console or an ambient mic), **the lock button** (lock every track) and **Patch** (fold the In and Out columns away).
- **The status line**: MIDI Bridge and the console, and your interfaces (click it for **Settings** → **Audio**); on the right, zoom and track height.

The session's files, the interfaces and the app's own things are in the **menus** and **Settings**, not the toolbar: **File** (New, Open, Open Recent, Close, Save, Save As, imports, Export, Remove Last Capture, Clean Up), **Edit** (Undo, Redo, cut, copy, paste, split, clip gain, mute and polarity), **Transport** (play, record, go to start, loop, **Locate…**, **Go to Marker…**, **Go to Region…**, previous and next, arming, Before You Record, Schedule…), **Console** (Sync from Console, Recall Scenes in Playback), **Markers**, **Track** (Add Track, Add Ambient Mic, lock, move, Rec-safe), **View** (the views, Lock, the side panel, zoom, heights, waveforms), **Window** (the meters, Minimise, Zoom) and **Help** (Keyboard Shortcuts, Quick Reference, Export Log, Send Feedback; macOS's menu search too).

Right-click a track for Rename, Colour, Move Up / Down, Lock, Rec-safe, Mono / Stereo and Delete; right-click a clip for Split, Clip Gain, Mute, Polarity, Lock and Delete.

On the right, the side panel: **Recording** (time left on the disk, the format, the record rate and Schedule), **Markers & regions**, **Patch**, **Console scenes**, **Before you record** (anything that would stop a take from being what you meant), **Interface Inputs** and the **FOH meter**. Click a section's title to fold it away and again to bring it back; each stays as you left it.

Interface Inputs shows a meter for every input of the multitrack interface, whether a track records it or not; an input no track records is dimmed, and pointing at a meter names its track. A live input with no track, or a track whose input is dead, stands out before Record.

The **View** menu shows or hides each section; **Show All Sections** brings every one back. Any section can have a window of its own: **View** → **Open in Own Window**, or the button at the top right of the section (it appears when you point at the section). The window is live, its controls work, and Space, `Ctrl+Opt+Space`, Home and `Cmd+S` work in it too. Its **pin** (top right) keeps the window above other apps, on every Space, as the Meters window's does; off, it's an ordinary window. Close it and the section goes back into the panel. Sections left in their own windows open again at the next launch, where you left them. If every section is hidden or in its own window, the panel closes up and the timeline takes the width. **View** → **Side Panel** (`Opt+Cmd+S`, or **the side panel button**) hides the whole panel at once and brings it back as it was; sections in their own windows stay open.

Settings (`Cmd,`, the toolbar, or **Show Recorder** → **Settings…**) holds the app's preferences, on tabs: **Audio** (the interfaces, Sub-Mixer and Sync's clock, the buffer, and **Look for interfaces again**), **Recording** (show length, auto-lock, splitting files by size, pre-record, a second copy on another drive), **Sync** (Record when timecode runs and when it stops, Timecode from, Follow timecode in playback, Play timecode on the LTC row's output, Session start timecode; those marked *(this session)* are saved with the session), **MIDI**, **Console** (Sync from Console's offset), **Companion & OSC**, **Export**, **Meters**, and **App** (Keyboard Shortcuts, Licence, Check for Updates). Nothing changes until **OK**; **Cancel**, Escape or a click outside leaves everything as it was. The interfaces and the buffer can't change during a take: if you change them then, the rest is saved and Settings says so. A session's format and record rate stay in the Recording panel.

Every button tells you its key when you hover over it. `Ctrl+Opt+K` shows every shortcut.

2. Sessions

A session is a folder holding the tracks, the patch, the mix and every take's audio. The folder's name is the session's name. Inside it, the session file has the same name and ends in **.showrec** (for example **Arena Night/Arena Night.showrec**). Double-click it in the Finder to open the session. Keep the file in its folder: the audio is found from there. Sessions saved before 0.2.0 have a **session.json** instead; they open as before, and the next save replaces it with the **.showrec**. Sessions are offered a place in **~/Music/Show Recorder**, named with the date, and can go anywhere you choose.

- **New Session** (`Cmd+N`) can start empty, from **last night's session** (its tracks, patch and mix, but no audio), from a **template**, or from the console (section 15).

- **Open Session** (`Cmd+O`) lists the saved sessions, with recent ones marked. **Other…** opens one from anywhere: choose its **.showrec**, or its folder. If audio a session uses is missing, it says which.
- **File** → **Open Recent** lists the last ten sessions opened or saved, newest first. One that's no longer on disk is greyed. **Clear Menu** empties the list.
- **Save** (`Cmd+S`). A dot after the session's name means there are unsaved changes. The first save of a new session asks for its name and where it goes. If you press Record, import audio or export first, the session saves by itself under its dated name in ~/Music/Show Recorder, so a take never waits on a dialog. The session is also saved by itself at the end of every take. An untitled show saved that way still asks for its name at your first **Save**, offering the dated name it was saved under: keep it, and it stays where it is, named.
- **Save As…** (`Shift+Cmd+S`) saves the session as a new folder, under the name you type, wherever you choose. The window carries on in the new one. The old folder stays as it was at its last save. If the session has audio, choose:
  - **Copy Audio** copies the recordings and imported files into the new folder, so it stands alone. A whole show can take a while; progress shows in the status line.
  - **Leave in Place** makes the new session play the audio from the old folder, so keep that folder where it is.

Save As waits until a take, an export or the take's live mix has finished. If a take starts while the audio copies (timecode, MMC or the keypad), the take wins. The copy stops and the half-made folder is removed. The session stays in its old folder, and you can Save As again after the take.

- **Save as Template…** keeps the tracks, patch and mix to start a new session from.
- **Import Mix Settings…** takes another session's mix (levels, pans, buses, compressors and the mastering chain) onto this one's tracks, matched by name. It reports anything it couldn't match. One `Cmd+Z` undoes it.

Undo and redo (`Cmd+Z`, `Shift+Cmd+Z`) cover every track, patch, clip, marker and mix edit. A fader drag or a stretch of knob turning is one step.

Pre-record (Settings → Recording): **Off**, **2 s**, **5 s** or **10 s**. While the interface is open, Show Recorder keeps the last few seconds of every input, armed or not, and each take starts with them, so a count-in, or a downbeat you pressed Record late for, is in the files. The take and its clips start that much before you pressed Record; the Recording panel's detail line shows the setting. Just after the interface opens (or the setting changes) a take starts with as much as has been kept so far, and never before the timeline's zero. Tracks armed during a take start where they're armed. It holds memory while the interface is open, shown beside each choice: every input × (the seconds + 2) × the rate × 4 bytes (16 inputs at 48 kHz, 5 s: about 21 MB). It can't change during a take.

Second copy to (Settings → Recording): every take written twice, the second time to a folder you choose on another drive (**Choose…**), in a folder named after the session: *Drive B/Night 1/audio/Take 001*, with the session file beside it, so if the main drive dies the copy opens as the session itself, takes and all. The copy has its own buffer and its own writers: a second drive that is slow, fills up or is unplugged mid-take never touches the main recording. The Recording panel shows the copy's drive, its free space and time left, and during a take whether it is losing anything; if it isn't connected at Record, the take goes to the main drive alone and the panel says so. A copy on the same drive as the session is allowed but warned about: it won't survive that drive failing. If the copy comes out incomplete, Stop says so (the take itself is whole). It costs twice the recording memory, about 0.5 MB a track.

Split files (Settings → Recording): off, each track's take is one file, which grows past 4 GB as RF64 (most tools open it). Every 2 GB or every 4 GB, a take's files are split into parts, numbered -001, -002 and so on, every track at the same sample, and each part is a clip on its track, end to end with the next. The live mix needs one file per track: with splitting on, the mix is exported at Stop instead, and during the take the tracks show the red recording band rather than a growing waveform.

Backups. Every save also leaves a timestamped copy of the session file in the session's **backup** folder, at most one every five minutes, the newest 50 kept. Each backup is a **.showrec** named with the session and the time. If the session file is ever damaged, copy the newest backup into the session's folder in place of its **.showrec**, renamed to match, then open the session. A backup can't be opened from the backup folder itself. Backups hold the session, never the audio.

Quitting, closing the window or starting a new session with unsaved changes asks **Save / Cancel / Don't Save**. If a save fails, the app stays open and says why.

3. Tracks and Sub-Mixer and Sync

Above the inputs is the **Sub-Mixer and Sync** section. Its rows are kept apart because the show needs them for more than one thing:

Row	What it is
L/R Mix	The console's main mix: what Find Songs reads and what the FOH meter shows when you're not playing back.
Desk mixes	Any auxes, groups, matrices or mains you record here rather than among the inputs (see **Add Track**). Out of the export mix until you unmute them in the Mixer.
Ambient Mics	The crowd mics, summed on the Ambient Bus in the export mix. Each row is usually an **aux** of a crowd mic recorded among your inputs (its **In** choice, as Click's), so your patch stays in order and nothing records twice; it can record its own input instead. **Track** → **Add Ambient Mic** adds a row, unpatched.
Click	The click. It can record its own input, or be an aux of another track. It's out of the export mix until you unmute it in the Mixer.
LTC	Timecode. It can record its own input, or be an aux of another track.

Click, LTC and the ambient mics as auxes. A click, timecode or crowd mic recorded among your inputs (so your patch stays in order) can still feed its row: the row's **In** choice, **Off**, **Own input** or **Aux of** a track, plays that track's audio on the row without recording it twice. The choice is in the track list's **In** column (fold the patch columns away and it's hidden) and beside the row's name in the **Patch Matrix**.

You patch these rows yourself: a new session leaves them all unpatched, and nothing (Sync from Console, New Session from the console, cascading a selection) patches them for you. Their playback goes only to the outputs you give them, never to their inputs' channels. An unpatched row records nothing and isn't a problem in **Before you record**.

Below them are the **inputs**: one track per channel or stereo pair. **Add Track** adds one, or as many as **How many** says (on consecutive inputs; a name you give is numbered, "Vox 1", "Vox 2"…; mixes are numbered on, Aux 5, Aux 6…; one undo), on the next free input: **an input channel** (mono or stereo), or **a mix from the console**, an aux, group, matrix or main that the desk sends out on that input. A mix track is stereo when the mix is, carries a small tag (AUX 3, ST GRP 1…), and **Sync from Console** names it from that mix on the desk rather than from an input channel. **New Session** → **From the console** also lists the desk's named mixes, unticked: tick those you want to record. Where the desk doesn't say whether a mix is mono or stereo (the DiGiCo SD's auxes and groups, the Wing's buses), its row has a **stereo?** tick, ticked: untick it for a mono mix. **Record in** says where a mix goes, for Add Track and New Session alike: **the Sub-Mixer and Sync section** (the default), as a row of its own under L/R Mix, unpatched until you patch it to the input the desk sends it on; or **the multitrack**, among the inputs, on the next free input (after the channels, from the console). Show Recorder remembers the choice. A mix's right-click menu has **Move to Sub-Mixer and Sync** or **Move to the Multitrack**; it keeps its patch and clips. Every channel from the desk is an input, a click, LTC or crowd mic included, so your patch stays in the desk's order. Each track has:

- a name (double-click to rename) and a colour;
- an **arm** button. Only armed tracks record;
- a **mono/stereo** switch. A stereo track records two files, L and R;
- its meter: the input, live before and during a take; in playback, what's on the track, whether or not the track is in the mix (the Mixer's meters show what it adds to the mix). An **armed** track shows its input in playback too, so you can watch the console's mix come back while the multitrack plays into it.

Select tracks with a click, `Shift`+-click or `Cmd`+-click. **Delete** removes the selected tracks. **L** locks the selected tracks (or clips) against edits.

Reorder by dragging a track's number (an ambient mic's grip) to where it should go, or select tracks and press `Cmd`+↑ / `Cmd`+↓. Inputs move among the inputs and ambient mics among the ambient mics; L/R Mix, Click and LTC stay where they are. A track keeps its patch and mix wherever it goes.

Arm all: click **Arm** at the top of the arm column (again to disarm all), or `Ctrl+Opt+A` / `Ctrl+Opt+D`. Rec-safe tracks keep what they are.

A narrower track list: click **Patch** in the list's header (or `Ctrl+Opt+I`, or **View** → **Patch Columns**) to fold the **In** and **Out** columns away; the soundcheck **M** and **S** stay. Click it again to bring them back. The patch is still in the **Patch Matrix** and the side panel's **Patch** section, and Show Recorder remembers the choice.

Rec-safe: `Opt`+-click a track's arm button. An amber ring marks it. A rec-safe track can't be armed or disarmed, its input patch and mono/stereo can't change, and it can't be deleted. A cascade or an unpatch drag across it passes it by. Its outputs stay free: they're playback, not the recording. `Opt`+-click again to release it.

4. Patching

Each track records from one input on the Multitrack interface (two for stereo) and plays back to one output.

A track patched past the interface's inputs (input 9 on an 8-input interface) records nothing: its input shows **red** in the track list, the Patch section says so, and if it's armed, **Record** names it when the take starts. The take runs for every other armed track.

Settings → **Audio** → **Look for interfaces again** finds one just plugged in, or an Aggregate Device just made in Audio MIDI Setup. Two interfaces with the same name (macOS calls every new one "Aggregate Device") are listed separately, numbered, and Show Recorder opens exactly the one you chose, by its system ID, even after a relaunch. An interface with only inputs (a built-in or USB mic) records; one with only outputs plays back and records nothing.

- **Cascade**: select tracks, type the first channel, then **Inputs** or **Outputs** to send them to consecutive channels from there.
- **Playback out where it came in** sets every track's output to its input: the virtual soundcheck patch.
- **The Patch Matrix** view is the patch at a glance: tracks down the side, the interface's channels across.
  - Click to patch, or drag down across tracks to patch them in order.
  - `Shift`+-drag the L or R of a stereo pair to move one side. Drop one side onto the other to swap them.
  - `Opt`+-click, or `Opt`+-drag, unpatches.
  - A channel recorded twice lights amber. Channels the interface doesn't have are hatched.
  - Dashed shadows show where a track plays out (on the Inputs page) and where it records from (on the Outputs page).

A second interface (Sub-Mixer and Sync out) can play the L/R Mix, Ambient Bus, Master, Click and LTC, in sync with the multitrack. Set its **Clock**:

- **Shared**: the two interfaces share word clock, or one Dante grandmaster.
- **Separate**: each interface runs on its own clock and the engine corrects the drift.

The engine pill shows that interface's rate, latency and drift.

5. Recording

Arm the tracks, then **Record** (`Ctrl+Opt+Space`). Every armed track records to its own file in the session's **audio/Take NNN** folder, as Broadcast WAV stamped with where it starts. Each file is named with a number and the track's name: the **Inputs** first, from **001**, in their order in the track list, then the Sub-Mixer and Sync rows (L/R Mix, Ambient Mics, Click, LTC): *001_Kick.wav*, *002_Snare.wav* … *033_L_R_Mix_L.wav*. Exported stems are numbered the same way. **Stop** (or Space) ends the take. At Stop, the take is placed on its tracks and the session is saved.

Arming during a take. Arm a track while recording and it starts recording there, in its own file, placed where it started. Disarm one and its file finishes there. Arm it again and it records a new file (*-a2*, *-a3*…), each its own clip, so nothing is overwritten. A track needs its input patched before the take starts to be armed in it, and the patch and the format still can't change during a take. If **Render the mix during each take** is on, an arming change pauses it and the mix is finished at Stop, with every track as recorded.

Format (the Recording panel): **WAV 24-bit**, **WAV 32-bit float**, **FLAC 24-bit** or **FLAC 16-bit**. WAV (Broadcast WAV) is what every DAW reads. FLAC is lossless and smaller: busy channels take 60–75 % of WAV's disk, quiet ones and

silence far less, and the disk time left counts 75 % to stay on the safe side. A FLAC file carries where it starts in a TIME_REFERENCE tag, which Reaper and most DAWs read; one that doesn't places it at zero. Takes play, edit, export and recover in FLAC as in WAV.

Record rate (the Recording panel): the rate takes are written at, from 44.1 to 192 kHz, whatever rate the interface runs at. Show Recorder never changes the interface's rate: it converts between the two as audio comes in and goes out, so a 96 kHz interface can record 48 kHz files at half the disk. The panel shows both ("Interface 96 kHz, recording 48 kHz (converted)"). Converting is the heaviest work Show Recorder does, so while it converts, it opens only the interface's channels the session uses (up to the highest input and output any track or bus is patched to) rather than all of them: on a 128-channel interface such as Waves SoundGrid that's the difference between a quiet show and constant overloads. **Interface Inputs** then meters only those, and says so. At the interface's own rate every channel is open. With a dLive (96 kHz), recording at 96 kHz converts nothing. **Above the interface's rate** (96 kHz files from a 48 kHz interface), every channel is converted up as it records: it costs CPU on every track, at many tracks enough for dropouts, and adds nothing to the sound. Show Recorder warns when that starts (the session opening, the interface chosen, the interface's rate changing, as when a Dante network is switched) with **Record at** the interface's rate, keeps an amber chip in the transport row while it lasts, and says so again at Record. Recording **below** the interface's rate is a choice and isn't warned about. Changing a session's rate keeps its clips, markers and regions where they were; a new session starts at the last rate you chose. Audio at another rate (an import, an old take) is converted as it plays.

Buffer (**Settings** → **Audio**): the interfaces' buffer, 1024 frames unless you choose 512 or 2048. Show Recorder monitors nothing, so a bigger buffer costs nothing you hear and gives more margin against dropouts on a busy Mac. Neither the rate nor the buffer can change during a take.

The Recording panel works out how long the disk can record at the armed tracks, rate and format, against the **Show length** set in **Settings** → **Recording**. It warns you:

- when less than the show length fits;
- when a FAT32 drive's 4 GB file limit would cut a track short;
- when the session is on a network volume.

Whenever there's a warning, a badge shows in the header.

Before you record lists what would make the take wrong: an unpatched armed track, a channel recorded twice, the interface missing.

Waveforms draw while you record, a second or so behind, and are complete the moment you press Stop, even after a long show.

A take is never lost to a crash. If the app or the Mac stops mid-take, the take is recovered when the session next opens, up to the last moment written to disk. Quitting mid-take asks first.

Timecode. Patch the **LTC** row to the input carrying timecode and the header shows the incoming code, its rate and its state: green when locked, amber while it latches or rides a dropout. Every take notes the timecode it started at (in the take report and the session). Show Recorder remembers the last rate that locked, so the next lock takes about half a second.

A timecode timeline. The first take, or the first playback following timecode, that sees code sets **Session start timecode** (**Settings** → **Sync**) to the code's hour: 20:13 code makes 20:00:00:00. From then on the clock reads timecode (the session time is its tooltip), and every take started while code runs is placed at its code, to the sample, so each night of a run lines up with the others. Code earlier than Session start timecode moves it back an hour and the session with it. Change Session start timecode to relabel the timeline: nothing moves. Empty it and the session isn't a timecode session.

Follow timecode in playback (**Settings** → **Sync**, or the transport row's **LTC** button) is virtual soundcheck driven by the show: when timecode starts, playback rolls from where the code says, follows it wherever it goes, and stops when it stops. It never runs during a take.

Timecode from (**Settings** → **Sync**) is what the session follows: **LTC** on the LTC row's input, or **MTC** on Show Recorder MIDI In (see below). Everything above follows it; placing a take to the sample is LTC's (MTC places it to the frame).

Play timecode on the LTC row's output (**Settings** → **Sync**) sends timecode in playback on the LTC row's output, generated from the timeline (Session start timecode, or 00:00:00:00 without one), so a desk or a show controller can follow Show Recorder in virtual soundcheck. It starts and stops with playback, follows every locate, and is silent during a take.

Record when timecode runs (**Settings** → **Sync**, the transport row's **AUTO** button, or **Transport** → **Record When Timecode Runs**; saved with the session) starts a take when the code locks, and stops it when the code has been gone for 5 seconds. The setting below it sets how long, or **Keep recording**: the code starts a take and only you stop it. Short dropouts ride through. A take you start yourself is yours: timecode stopping never ends it. If you stop a take while the code is still running, it isn't restarted until the code stops and comes back. If a take can't start (nothing armed, for example), the Recording panel says why, once.

Space during a take only stops it. It never starts playback over a recording.

During a take:

- the Mac is kept awake for as long as the take runs (closing a laptop's lid still sleeps it);
- New, Open, Import, Export and Remove Last Capture are greyed out in the File menu.

Lock (at the right of the header) puts a veil over the window. The take, the clock and the meters carry on underneath, but no key or click gets through. To unlock, hold Escape, or hold **Unlock**, for a second. **Auto-lock** (**Settings** → **Recording**) locks the window by itself after 1, 2, 5 or 10 minutes without a key or a click during a take.

Schedule… (the Recording panel) starts recording every armed track at a time of day, and/or stops at another: each the next time it comes round, once. The panel says what's scheduled. Keep Show Recorder open and the Mac on; a Mac asleep at the start time records as soon as it wakes.

Meters (`Shift+Cmd+M`, or the Window menu) shows REC and the take's time, the session time, time left on the disk, dropouts, the FOH meter, and a meter for every track that records: armed ones lit, in the desk's colours, so a dead input on an armed track stands out. Each track's name runs up beside its meter, so many fit side by side. `Shift+Cmd+M` shows or hides them wherever they are, and they come back at launch as you left them. They go in one of three places: their **three-dot** menu (**Meters go**) or the Window menu chooses.

- **Docked top** or **Docked bottom**: a strip across the timeline, above the ruler or below the tracks, with the meters in one row. Drag the handle on the strip's edge (the one facing the tracks) to make it taller or shorter, or choose **Strip height** (S, M, L, XL) in its **three-dot** menu; **Track meters** there sets how wide the meters are. Docked, the strip leaves out the clock, since the main window shows it.
- **Own window**: a small window of their own. With the **pin** (top right) on, the window stays above every other app, on every Space, and never takes the keyboard: put it over the console's screen or the DAW. With the pin off, it's an ordinary window that goes behind others. **Window** → **Keep Meters Window on Top** does the same. Closing the window hides the meters.

The track meters fill the window's height: make the window taller and they grow with it. A stereo track has two meters side by side, L and R. The input tracks come first, then the Sub-Mixer and Sync rows after a line; an aux row (a Click or LTC that plays another track) shows that track's level, and a row set to Off has no meter. The **three-dot** menu at its top right sets how wide they are (**S**, **M**, **L** or **XL**; at L and XL track names show in full) and how FOH is shown: **Bars**, two thin L/R bars across the top, or **Loudness & phase**, the main window's FOH meter beside the tracks: the average level as a bar with the peak held as a dot, momentary, short-term and integrated loudness, true peak, and the phase correlation needle. Loudness and phase are measured in playback, and on the L/R Mix input with a current audio engine; otherwise the meter says **PEAKS ONLY**. Integrated loudness and true peak run from the main window's **Reset**. Both choices are remembered.

Clean Up Session… (the File menu) moves every take and file in the session's folder that no clip uses into its **dead** folder: takes you removed, imports you deleted. Nothing is deleted; empty the dead folder in the Finder when you're sure. It lists what will move, and its size, first. A take still marked as recording isn't touched (it's recovered when the session opens). If an undo brings back a clip whose file was moved, the file comes back too.

Remove Last Capture (the File menu) is for a false start. The newest take's clips come off the timeline, and its folder moves to the session's **dead** folder: no audio is deleted. `Cmd+Z` brings the take back, folder and all.

6. Dropouts

The dropout counter, a chip in the transport row, counts anything that cost the recording audio:

- the disk falling behind, or refusing to write;
- playback streaming falling behind;
- the Sub-Mixer and Sync interface falling behind;
- the interface stopping sending audio. The counter blinks until it comes back. After 2 seconds Show Recorder starts the interface again, every 2 seconds until it answers, so a take carries on in the same files: after macOS's audio restarts, or a cable is pulled and put back. The time it was gone is a dropout, marked where it happened.
- the audio engine skipping audio it was handed;
- **the rate conversion falling behind** (recording at another rate from the interface's, the incoming audio is converted on its own thread, 50 ms ahead): its frames are silence;
- **WRONG INTERFACE**, in red: the take's audio is on another device than the one chosen. macOS can move a stream to the Mac's own mic or speakers when its audio restarts. The take gets a warning marker there, Show Recorder reopens your interface, and the alarm clears when the audio is back on it. After any restart of macOS's audio (a driver installing, waking from sleep), Show Recorder reopens your interface at once, under a take that's recording;
- **an overload**: the interface itself reporting that a buffer was late (macOS's own report). When overloads keep coming (10 or more in 10 seconds), the counter adds **what helps?**; click it for what to try, most effective first: **record at the interface's rate** (when the two differ, the conversion is usually the cause), **turn Pre-record off**, **a larger buffer**, and quitting other apps on the same interface. The log notes overloads once every 10 seconds with their count;
- **a take that isn't moving**: the interface is running but the take's time has stood still for a second, so nothing is being recorded. The counter turns red and reads **NOT RECORDING** (in the Meters too) until the take moves again. Stop and start the take; if it happens again, change the buffer in **Settings** → **Audio** and send the log.

Each dropout during a take is also a warning marker on the timeline, in yellow on the lower row of the markers lane (your markers are red and the desk's scene recalls orange, on the top row), where it happened: to the audio block for the disk, playback streaming, skipped audio and overloads, which the engine reports with their own position. Click the counter for the details and **Reset**, or its **×** to clear the count at once (the Meters' **Dropouts** line does the same when clicked). The the warning sign markers stay on the timeline. "Interface stopped" and **NOT RECORDING** have no ×: they clear themselves when the interface or the take recovers. Other notes have a **×** too: the rate warning (hidden until the rates change again) and any message in the bottom bar.

Load (a chip in the transport row): how much of each buffer period the audio engine uses, over the last second, and the worst since the take started (or the counter was reset). It turns amber from 70 % and red from 90 %: near 100 % the interface misses deadlines, and a bigger buffer (**Settings** → **Audio**) gives more room. Its tooltip also counts the interface's overloads. **CPU** beside it is Show Recorder as a whole, every thread, as a percentage of one core: what recording, sample-rate conversion, FLAC or MP3 encoding, the waveforms and the window cost together. Load is the audio deadline; CPU is the total, and on a Mac with several cores it can pass 100 %.

7. The timeline

Beside the track list is the timeline: each track's clips with their waveforms, a ruler, and the **playhead**. Click the ruler or the lanes (or a marker, `[ ]`, Home) to move the playhead, stopped or playing; Play starts from it, and Stop leaves it where playback stopped. During a take the playhead is the take's and doesn't move.

The section bars (**Sub-Mixer and Sync**, **Inputs**) are rulers too: over the timeline they show the time, lighter than the top ruler, so it reads lower down a long session, and a click there places the playhead. Click a bar's name to fold its section away (its tracks still record).

- **Zoom**: `Cmd+=`, `Cmd+–`, `Cmd+0` (fit the show) and `Cmd+Shift+0` (the selection), or pinch, or `Opt+`-scroll at the pointer.

- **Track height**: `Ctrl+Alt+↑` / `↓`, `0` to reset, `F` to fit; the **–** / **+** beside the track height label in the status line; or `Cmd+Opt+`-scroll over the timeline. **One track's height**: drag its bottom edge; double-click the edge to put it back with the others. Fit and reset put every track back to one height. **Waveform size**: `Ctrl+Alt+=` / `–`, `9` to reset. All of these are in the **View** menu too, with zoom in, out, to fit and to the selection.
- **Folding**: click **Sub-Mixer and Sync** or **Inputs** above their tracks to fold them away and back. A folded section's tracks still record and play; the heading says how many are hidden.
- **Waveforms in dB** (`Ctrl+Alt+8`, or **View** → **Waveforms in dB / Linear**): the waveform's height in decibels, 60 dB from a lane's centre to its edge, instead of in proportion to the level. A quiet passage, room tone or a song's tail shows instead of a flat line. Switch back the same way.
- **Detail**: zoomed in, a waveform keeps its detail down to single samples, read from the audio itself, and on a Retina screen it's drawn a screen pixel at a time. While a zoom or a scroll is on its way, the waveform stays where it belongs and sharpens a moment later.
- **Select a time range** by dragging across the lanes. `Escape` clears it.
- **Clips**: click, or `Shift+`-click, to select. Drag to move: they snap to markers, regions, other clips and the cursor; hold `Cmd+` to drag freely. **N** turns snapping on and off. `Opt+←` / `Opt+→` nudge 10 ms (with Shift, 1 s). **Delete** removes them.
- **Editing**:
  - **S** (or **Edit** → **Split Clips at Playhead**) splits at the playhead: the selected clips, else the selected tracks' clips, else every clip the playhead is in.
  - Drag a clip's **edge** to trim it (the pointer becomes ↔); it can't go past the start or end of its audio.
  - `Opt+Cmd+`-drag inside a clip to **slip** it: the audio moves under fixed edges.
  - `Cmd+X`, `Cmd+C`, `Cmd+V` cut, copy and paste the selected clips. Paste lands at the playhead, back on the clips' own tracks, or on the selected tracks (the first copied track onto the first selected one). A mono clip can't land on a stereo track, or the other way round.
  - **Clip gain, mute and polarity** (the **Edit** menu): **Clip Gain…** (`Opt+G`) sets the selected clips' own level, –60 to +24 dB, before the track's fader; `Opt+↑` / `Opt+↓` nudge it 1 dB. **Mute Clips** (`Shift+M`) silences them and **Flip Clip Polarity** (`Opt+P`) inverts them; each toggles. They apply live while playing, without a click, and in every export. The waveform is drawn as it plays (bigger with gain, upside down when flipped, dimmed when muted) and the clip's name says **MUTED**, **Ø** and its gain. A split keeps them on both halves; copy and paste carry them.
  - Each is one undo step. A trim, a slip, a move and clip gain apply live, even while playing; a split, a paste or a new clip rebuilds playback for a moment, the playhead staying where it is.
- **Songs**:
  - `Shift+Delete` **ripple-deletes** the selected range (drag across the tracks), or the selected song (click it in **Markers & Regions**). It's cut out of every track, with its markers and regions, and everything after it moves back to close the gap. A region across the cut is shortened, not lost.
  - In **Markers & Regions**, point at a song and use its up and down arrows to move it earlier or later. It swaps places with its neighbour on every track, with its markers, and the gap between them stays between them, so moving a song and moving it back leaves the show as it was.
  - A locked track or clip refuses the whole edit and says which. Each is one undo step.
- **Locks**: **L** locks the selected clips (or, with none selected, the selected tracks). `Ctrl+Alt+L` / `U` lock or unlock every track.

A track's waveform takes its track's colour, including the desk's colours from Sync from Console.

8. Markers, regions and Find Songs

Markers and regions have their own lanes above the tracks.

- `Opt+M` adds a marker at the cursor; `Opt+Shift+M` adds a named one.
- `Opt+R` makes a region from the time selection.

- `Shift+E` makes the time selection from the selected clips, the first one's start to the last one's end, ready to loop, export or make a region.
- **Shift-click** on the lanes extends the time selection to where you click (from its far end); with no selection, it selects from the edit cursor. Dragging a selection past the edge of the window scrolls the timeline on.
- Drag either to move it. Double-click to rename. `[` and `]` step to the previous and next one.
- Orange markers are the console's scene recalls (section 15).
- **The Markers & regions list** (`Ctrl+Alt+M`) lists them all; click one to go there (a region also selects its time).

Find Songs (the Markers menu) makes a region for every song in the **L/R Mix**, split at its quiet gaps, ready to rename and export. You set:

- what counts as quiet (dBFS);
- how long a gap must be;
- how long a song must be;
- the padding either side.

Running it again replaces the regions it found before. Regions you placed or edited are kept.

9. Playback and virtual soundcheck

Play plays from the playhead; **Stop** and Space stop it; **Home** goes to the start.

Loop. Drag across the ruler to set a loop: a green bar along the ruler, on at once. Drag its ends to change it. **Loop** (`Shift+L`) switches it off and on again; off, the bar dims and the range is kept. Select a range on the lanes and press **Loop** to loop that instead. **Escape** clears the loop with the time selection. A loop plays when playback starts before its end; started after it, playback runs on. It is gapless and to the sample, and never applies to a take: Record runs straight through it.

The number pad is a transport too: **0** plays and stops (during a take it only stops), **3** records, **Enter** adds a marker, and **.** *n* **.** (or **.** *n* **Enter**) goes to the *n*-th marker.

Every track plays back out its patched output: patch the outputs to the console's inputs and the show plays back through the desk, a **virtual soundcheck**. Re-patching an output while it plays takes effect at once, without a gap; re-patching an input reopens the interface for a moment. Each track's **M** and **S** beside its output are the soundcheck's mute and solo: **M** stops that track's playback to the console, and **S** plays only soloed tracks to it. They're saved with the session and change nothing recorded, and nothing in the export mix (that's the Mixer's mute and solo). The export mix plays through the L/R Mix and Ambient buses to the master's outputs on the Sub-Mixer and Sync interface. Every Mixer and Processing change is heard at once.

10. The Mixer

The Mixer is the **export mix**: what Export writes as the stereo mix. It doesn't touch the recording or the console.

- **Strips** for every track that can feed the mix: level, pan, mute, solo, and which bus it goes to. Moving an input onto the **Ambient Bus** offers an **Ambient Mics aux** of it instead (**Add Ambient Aux**): a row in the Sub-Mixer and Sync section plays the input and feeds the Ambient Bus, while the input stays where it is in your patch, records once, and goes back to the L/R Mix bus, out of the mix. **Feed It Directly** keeps the input feeding the Ambient Bus from its own strip. **Click** has a strip too, muted: unmute it to put the click into the mix, as a reference in a recording or an export. LTC never goes into the mix.
- **L/R Mix** and **Ambient Bus**, each with a fader, mute and balance, and a **bus compressor**.
- **Master**: the fader into the mastering chain.

Each strip shows its processing as buttons, lit when in. Click one to open it; `Opt`**+click** (or `Shift`**+click**, as in Console Control) to switch it in or out without opening it (its settings are kept, and `Cmd+Z` undoes it).

Meters on every track, bus and the master. In playback they show the export mix. Recording or stopped, the L/R Mix and Ambient Bus meters show the Sub-Mixer rows' live inputs (the L/R Mix row, the ambient mics) through their faders, mutes and solos; the Master's meter stays dark, since live inputs aren't heard through the master. Faders and

pan are heard as you move them; each move is one undo step when you let go. The mix is saved with the session, carried into tomorrow's session from last night's, and into templates.

11. Processing: the mastering chain

The **Processing** view is the master's mastering chain, and each bus's EQ:

```
master:  sum -> mix EQ -> mix compressor -> tape -> clipper -> fader -> mix limiter
buses:   sum -> mix EQ -> compressor -> fader -> balance
```

Every stage is out until you switch it in. The **Chain** tab shows the whole chain and its stages' meters; each stage has its own tab.

Stage	What it does
EQ	Up to 24 bands, dynamic bands, zero latency or linear phase, with an analyser behind the curve. The graph spans 20 Hz to 20 kHz; **Freq** sets a band from 10 Hz to 30 kHz.
Comp	The mix compressor: optical and discrete thresholds, ratio, attack, recovery, a transformer stage and output.
Tape	Drive into tape saturation, and output.
Clip	A clipper: its ceiling, softness, input and output.
Limit	The mix limiter: eight styles, ceiling, attack and release, true peak, and dither for the export.

Each change is heard live, and each is one undo step. **Dither is for the export only**: live, the limiter never dithers. Whatever latency the chain adds, the export is lined up to the sample.

12. The FOH meter

The meter shows the master in playback. Stopped or recording, it shows the console's L/R Mix feed as it comes in, or any track you pick under it: a stereo track as its pair, or a mono track alone or against a second one, for loudness, true peak and correlation on any two channels.

Pre / Post (beside its title) chooses where it reads in playback. **Post**, the default, reads the master after the export mix's processing (EQ, compressor, tape, clipper, limiter): what an export sounds like. **Pre** reads the mix going into the master's processing: every track and bus summed, before EQ, compressor, tape, clipper, fader and limiter; loudness and phase read on Post. Stopped or recording, the meter reads the input either way: the processing runs on playback and export, never on what's coming in.

- **L/R bars**, Dorrough-style, in LED segments of 1 dB: the lit column is the **RMS** level (over 300 ms), a single lit segment above it the peak, held 1.5 s. **RMS** beside the meter reads it in dBFS. On an audio engine older than this version, the column is an average of the peaks and RMS reads —. Unlit segments glow faintly, as LEDs do. **PK** shows the highest peak; click it to clear it.
- **The scale** reads from your **0**, the alignment level (**Settings** → **Meters**: –18 dBFS unless you choose –20 or –14), from –30 up to digital full scale (+18 at –18 dBFS), marked at +18, +14, +10, +6, +3, **0**, –3, –6, –10, –20 and –30. Green below 0, amber from 0, red in the top 6 dB. Every meter in Show Recorder, the track meters, the Mixer's and the Meters, uses the same scale and colours, so amber always means above your 0.
- **M** momentary (400 ms) and **S** short-term (3 s) loudness.
- **I** integrated loudness and **LRA** loudness range since Reset (BS.1770-4, EBU Tech 3342).
- **TP**: the highest true peak since Reset (4x oversampled).
- **Match…** moves the master fader so the mix's integrated loudness, as measured in playback (**Post**) since **Reset**, lands on a target: –14, –16, –18, –23 or –24 LUFS. Play a representative stretch first. A limiter in the chain doesn't respond dB for dB, so play on and match again to settle it. One undo step.
- **Correlation**: +1 mono, 0 wide, below 0 out of phase.

Reset starts integrated, LRA, true peak and the held peak over.

In time with what you hear. In playback the sound reaches the speakers a little after Show Recorder plays it: the interface's buffer, and when the mix plays on a second device (Sub-Mixer and Sync), the handover to it, which can be a third of a second. Every meter, the playhead and the clock follow what you hear, not what's being played: they're held back by exactly that delay, measured from the interfaces themselves. Recording or stopped, the meters show the inputs as they arrive.

In its own window (**View** → **Open in Own Window** → **FOH Meter**), the meter fills the window and **Horizontal** turns it on its side: L and R across, the scale beneath. **Vertical** turns it back.

13. Export

Export (`Cmd+E`):

- **What to export**: **the stereo mix** (L/R and crowd, through the Mixer and the mastering chain), and/or **the stems** (every track with audio, as recorded, for another DAW).
- **Over**: the whole show, each song (each region), the time selection, or **the selected clips** (the first one's start to the last one's end). The dialog opens with the choices you made last time.
- **Format**: WAV 24-bit, WAV 32-bit float, or WAV 16-bit (dithered). Every file is stamped with where it starts. **Stereo stems as separate L and R files** if the other DAW wants them.
- **The mix as**: **WAV**, **WAV and MP3**, or **MP3** (128 to 320 kbps, encoded by LAME). MP3 is the one to send: a few megabytes a song. A session above 48 kHz is converted to 48 kHz for it, MP3 having no higher rates. The stems are always WAV.
- **Files**: **WAV** (Broadcast WAV: stamped with where it starts, so stems and songs land where they belong) or **AIFF** (no time stamp, so a DAW places it at zero; up to 4 GB a file, which is about three hours of 24-bit stereo at 48 kHz: choose WAV for longer).
- **Mix loudness**: **As mixed**, or normalized to **–14**, **–16**, **–18**, **–23** or **–24 LUFS**, never letting its true peak pass the ceiling you choose (–0.1 to –2 dBTP). This is gain only: if the mix's peaks won't let it get to the target, it stops at the ceiling and the report says how far short it is. The mix limiter (**Processing**) is what makes a mix louder than its peaks allow.
- **Folder name** and **Mix file name**: patterns with **{session}**, **{span}** (the song's name, **Show** or **Selection**), **{date}** and **{n}** (the song's number). The defaults are *{session} - {span}* and *Mix*. A name with an unknown token is refused before anything is written.
- **A CUE sheet of the songs**: with the whole show's mix, a *.cue* file beside it with a track for each region, at its start. CD tools and players read it to split the show into its songs.
- **Where**: the export folder (**Settings** → **Export**; the session's **Exports** folder unless you choose one), or a folder you choose for this export.

Export remembers these choices for next time.

Export runs in the background with progress and **Cancel**. Cancelling leaves nothing half-written. When it finishes, it gives each mix's integrated loudness, loudness range and true peak (and, normalized, the gain applied and where it landed), and **Show in Folder**.

The mix during the take. Turn on **Render the mix during each take** (**Settings** → **Export**) and Show Recorder renders the stereo mix (the L/R Mix and everything in the Mixer, through the mastering chain as you have it dialled) while the take records, a second or so behind it. At Stop it finishes in a moment and the file is in the export folder as *Session - Take 001 - Mix.wav*: the same file the Export would give you. Change the processing during the show and the mix follows from there, as heard.

It renders on its own, apart from the recording, so it can't cost the take a sample. If the take has a dropout, it pauses and finishes after Stop (taking as long as an Export of what's left). A take it can't follow becomes an ordinary export at Stop. Either way, the mix arrives.

Settings → **Export** also sets **The mix as** (WAV, WAV and MP3, or MP3) and the MP3's bitrate, for this mix and as Export's starting choice. The MP3 is encoded as the take records too, so it's ready at Stop with the WAV, however

long the show.

Uploading. Choose a Dropbox, iCloud Drive or Google Drive folder as the export folder and that app uploads each mix when the Mac is online. Show Recorder itself never goes on the network for it, and a mix only lands in the folder when it's whole, so nothing half-written is uploaded during the show.

14. Importing audio

Import Audio… (`Cmd+I`) brings in stems from another DAW: files, or a folder. It shows a review before anything changes:

- **Split-stereo pairs** ("Kick.L" and "Kick.R", also _L / _R, Left / Right, 1 / 2, and numbered exports such as "17 AMB L" / "18 AMB R") stay as mono tracks unless you tick **Join split L/R files into stereo tracks**, which shows how many pairs it found and is remembered. **Poly files** are split, a track per channel.
- **DAW numbering** ("01 Kick In") is removed from names.
- **Placed by their timestamps**, so stems line up as they did in the DAW.
- **Matched onto existing tracks** by name; the rest become new tracks. A stereo file named as the console's mix (**FOH Mix**, **LR**, **Main Mix**, **Stereo Mix**, **2mix**) goes onto the **L/R Mix** row, and a joined ambient pair onto **Ambient Mics**. Click and LTC files stay tracks among the inputs, so a sequential patch stays in order; choose the **Click** or **LTC** row in Goes to to put them there, or make the row an aux of the imported track.
- **Goes to** picks where each one lands: a new track, or any track of the same width (stereo onto stereo), the L/R Mix and Sub-Mixer rows included.
- Rename or leave out any of them in the review.
- **Copy into the session** (the default), or **use in place**; your choice is remembered.
- **The files' rate.** Files are never converted on import. Into an empty session at another rate, the review offers to make the session the files' rate; otherwise it says they will play converted.

Drag WAV files or a folder from the Finder onto the window to import them the same way. Dropped on a track's lane, a file goes onto that track at the time you dropped it, when its width fits: a stereo FOH mix onto the **L/R Mix** row, say. Dropped elsewhere on the timeline, the files become new tracks at that time, keeping their spacing; the review opens first either way.

Import Reaper Project… (the File menu) brings in a Reaper project's audio: each track with audio as a track (mono or stereo from its first file), each item as a clip where Reaper had it (its position, length and where in its file it starts), and, if you leave it ticked, its markers and regions. A file moved since the project was saved is looked for beside the project and in its folders. What Show Recorder can't place exactly is listed in the review and left out: an item playing at another speed, MIDI and other non-audio items, muted items, files with more than two channels, and files it can't find. Copy the files in (the default) or use them where they are; one undo step.

Imported tracks get the next free inputs, unarmed. The whole import is one undo step.

15. Working with the console

Show Recorder reads the console through **MIDI Bridge - dLive** (free, from dylanmaudio.com). Start MIDI Bridge and connect it to the console first; no Allen & Heath software is needed.

Sync from Console reads every input channel's name and colour and shows what would change before it changes anything:

- each input and ambient track takes its input channel's name;
- an **offset** for when the multitrack's inputs aren't the desk's channels 1:1 (also in **Settings** → **Console**);
- names stay unique; armed tracks can be renamed;
- each track's mix follows its new name;
- the desk's colour marks the track and its waveforms.

It's one undo step. It can't run during a take: reading names sends a few hundred queries to the console.

Scene markers. With MIDI Bridge running, every scene the console recalls during a take becomes a marker where it happened, named after the scene ("Scene 12: Verse 2"; on the DiGiCo SD, by the number the desk shows, "Snapshot 1.93: Verse 2"), in orange on the markers lane. The **Console scenes** panel says whether the link is up. Scene markers are markers like any other: move, rename or delete them.

Recall scenes in playback (the Console scenes panel) is for virtual soundcheck. In playback, Show Recorder recalls on the console the scene of the last scene marker before the play cursor, whenever that changes: when playback passes a scene marker, when you press Play, and when you jump. Jump into the middle of the second act and the desk is in the second act's scene. It drives the console, so it's off until you turn it on, and it never runs during a take. It works on the dLive, the SQ, the Behringer Wing and the DiGiCo SD. On the Wing a scene's number is its place in the open show's list, so reordering the list between the take and playback, or having another show open, makes a marker recall a different scene; with no show open, nothing is recalled. On the DiGiCo SD (with MIDI Bridge 1.4.111 or later) a marker keeps the snapshot's number as its screen shows it, 2.00 or an inserted 1.50, and recalls that snapshot wherever it has moved to in the list; if the desk no longer has that number, nothing is recalled, never another snapshot, and the log says why. With an earlier MIDI Bridge the SD recalls by place, as the Wing does. **A session remembers which console its scene markers came from**, and recalls only on that kind of console: a show recorded on an SD, played back with MIDI Bridge on a dLive, recalls nothing there (the log says MIDI Bridge refused it) rather than the dLive's scenes of the same numbers. Sessions whose markers were made before 0.2.15, or with a MIDI Bridge older than 1.4.92, don't know their console and aren't protected this way. A Wing show opened with no scene loaded makes no marker. Re-recalling the scene the desk is already on may not make a marker on the Wing or the SD.

New Session → **From the console** builds the session from the desk:

- a track for every named input channel, in channel order;
- stereo where the names say so: "Keys L" and "Keys R", "OH Left" and "OH Right", "Amb SL" and "Amb SR", or "Gtr" and "Gtr R" become one stereo track;
- the crowd mics as Ambient Mics, and Click and LTC on their rows;
- each recording from its channel, less the offset, with the desk's names and colours.

It shows the session before it's made, and you choose there:

- **Record console channels** *from* **to** *to*: only those (for example 1 to 52, what the desk sends to your multitrack). **L/R Mix** records from the two inputs straight after them.
- Each row in or out (its tick).
- **stereo** or mono for each two-channel row. The desk doesn't say which of its inputs are stereo, so a channel whose next one still has the desk's default name, or the same name, is offered as **stereo?** (in amber): that's how a stereo input looks through the desk. Untick it if it's really mono. A mono track on a stereo input records half of it, which only shows when you play it back; a stereo track on a mono one just records a silent right side. On the DiGiCo SD and the Wing, a stereo channel is one channel number carrying two signals, and the desk says which ones are stereo: such a channel is offered as **stereo** (ticked), and its left and right go on two inputs side by side, so every channel after it comes in one input later (the desk's outputs patched in channel order). Untick it to record its left only; the channels after it stay where they are.

Sync from Console names a stereo track the same way (the pair's name, "Keys", not "Keys L").

Companion. With Bitfocus Companion on this Mac and the dylanmaudio module, a deck can **Record**, **Stop**, **Play / Stop playback**, **Go to start**, **Add marker** (named or not), **Go to marker** *n*, step to the **previous** and **next** marker, **Reset** the dropout counter and **Show** the window. It can show the transport, the time, the take, how many tracks are armed, the dropouts, the recording time left, a disk warning, the session and unsaved changes. Companion can't move the transport during a take, and Play / Stop never stops one. **Settings** → **Companion** has two switches: **Allow Companion control** (on), and **Lock show-critical controls** (off), which stops a deck from ending a take. The transport row's **Companion** chip says whether it's running and how many decks are connected.

OSC. Any OSC controller (TouchOSC, Lemur, a lighting desk, Companion's generic OSC module) can do what a Companion deck does. Turn it on in **Settings** → **Companion & OSC**: **On, this Mac only**, or **On, from the network** for a tablet or a desk on the show network. It listens on port 9001 unless you change it. Send:

- `/sr/record`, `/sr/stop`, `/sr/play_stop`, `/sr/start`
- `/sr/marker` (with a name, or none), `/sr/go_marker` *n*, `/sr/prev`, `/sr/next`
- `/sr/reset_dropouts`, `/sr/show`

Send state to (*host:port*) sends the transport, the time, the take, the armed tracks, the dropouts and the disk time left as `/sr/state/<key>` whenever they change. **OSC Addresses…** lists every address to copy into a controller. OSC works under the same two switches as Companion: with **Lock show-critical controls** on, `/sr/stop` can't end a take.

MIDI (**Settings** → **MIDI**; the transport row's **MIDI** chip shows what's on, and its tooltip any problem):

- **MTC out** adds a MIDI source, **Show Recorder MTC Out**: in playback it sends the timeline's timecode (a full-frame message at every start and locate, quarter frames while it rolls). Never during a take.
- **Show Recorder MIDI In** is always there, and **Also listen to** adds one MIDI source. Left as it comes, that's **MIDI Bridge's Return port**, which passes on what the dLive sends; Show Recorder finds it whenever MIDI Bridge is running, even if it starts later. Choose **Nothing else** to hear only Show Recorder MIDI In, or any other source. MTC on either is what **Timecode from: MTC** follows.
- **Follow MMC**: MMC from the desk plays, stops, pauses, records (Record strobe) and locates (with Session start timecode set). **Rewind** and **Fast Forward** wind the playhead until the desk's next command: 10× for 2 seconds, 30× to 6 seconds, then 120×; Play plays from where it got to, Stop or Pause leaves it there. Never during a take. Off until you turn it on.
- **MMC Stop may end a take**: off, an MMC Stop during a take is refused and the panel says so; on, it stops the take. Stopping a take is show-critical, so it's yours to allow.

16. Beta, licence and updates

The beta runs until **30 November 2026** (0.2.1 on; the betas before it end on 31 October). Until then everything works, on any number of tracks. The end date is on the window from the first launch, and the last two weeks warn at every launch. After the end date, a beta still opens, plays, edits and exports your sessions, but starts no new recording: install the current version to carry on. The date is checked once, at launch; a session that runs past midnight on the last day keeps recording.

The licence (from the release): without a licence a take records up to **8 tracks**, and everything else works in full. A licence unlocks any number of tracks, on two Macs, and covers **this and every future version, with no upgrade fees**. **Licence…** (the app menu, or Settings → App) activates and deactivates. Activating is the only time Show Recorder goes online, and never while recording or playing. A licence is never checked during a take: a take that started is never stopped.

Check for Updates… (the app menu, or Settings → App) asks once, when you choose it, and never by itself or while recording or playing. A beta hears of the next beta.

17. Logs and support

Every launch writes a session log to **~/Library/Logs/dylanmaudio/Show Recorder/**: the interfaces and their rate and buffer, every take started and stopped, every dropout and error.

- **Export log** (the footer) puts a diagnostics zip on your Desktop: this launch's log, the one before and your settings. Send it with any report; nothing leaves your Mac unless you send it.
- **Send feedback** (the footer) sends a short message straight to me.
- **Quick Reference** (the footer) opens the two-page guide.

18. Keyboard shortcuts

Show Recorder's keys are the family's: Console Control's where the two do the same thing (since 0.2.7). Single keys are each app's main job, `Shift+` + a key is its pair, `Opt+` is markers and nudging, `Cmd+` is the app, and `Ctrl+Opt+` is the console and the view. Only the `Cmd+` and `Ctrl+` keys show in the menus: the others would take letters from

text fields, so they work in the window and show on the shortcut sheet and in the tooltips.

Change any of them in **Keyboard Shortcuts** (`Ctrl+Opt+K`, **Help** → **Keyboard Shortcuts…** or **Settings** → **App**): a tab for each kind (Transport, Session, Markers, Edit, View, Console, App), and a field for each command. Click the field and press the new keys; `Delete` clears it, **the reset button** puts its default back. A key another command has turns red and Show Recorder names both; OK saves every change together, so two commands can swap keys. The menus and the buttons' tooltips follow at once. **Reset All Shortcuts…** (on the App tab) puts every default back. Keys the system keeps (`Cmd+Q`, `Cmd+W`, `Cmd+H`, `Cmd+M`, `Cmd+C`, `Cmd+V`, `Cmd+X`) can't be used, and Escape, the number pad and the mouse gestures don't change: they're on the **Fixed** tab. Your keys are kept with the settings and survive updates; a key you changed stays yours when the defaults move.

The defaults:

Key	What it does
Space	Play / stop (during a take: stop only)
`Ctrl+Opt+Space`	Record (`Cmd+R` until 0.2.6)
Home	Go to start
T	Locate: go to a timecode or a time
G / `Shift+G`	Go to a marker / a region
`Shift+L`	Loop the selection
`Shift+A`	Arm / disarm the selected tracks
`Ctrl+Alt+A` / `D`	Arm / disarm every track
`Cmd+=` / `Cmd+–` / `Cmd+0`	Zoom in / out / to fit
`Cmd+Shift+0`	Zoom to the selection
`Ctrl+Alt+↑` / `↓` / `0` / `F`	Track height: taller / shorter / reset / fit
`Ctrl+Alt+=` / `–` / `9`	Waveform size
`Ctrl+Alt+8`	Waveforms in dB / linear
`Ctrl+Alt+I`	The track list's In and Out columns (`Ctrl+Alt+P` until 0.2.6)
`Ctrl+Alt+X`	The Mixer (again: back to the tracks)
`Cmd+Shift+L`	Lock the window (hold Escape to unlock)
Pinch, `Opt+`-scroll	Zoom at the pointer
Escape	Clear the selection
N	Snap on / off
`Opt+←` / `Opt+→`	Nudge 10 ms (with Shift: 1 s)
S	Split clips at the playhead
`Opt+G`	Clip gain…
`Opt+↑` / `Opt+↓`	Clip gain up / down 1 dB
`Shift+M`	Mute / unmute the selected clips
`Opt+P`	Flip the selected clips' polarity
`Shift+Delete`	Ripple delete the selection or the selected song
`Cmd+X` / `Cmd+C` / `Cmd+V`	Cut / copy / paste clips
L	Lock / unlock the selected clips, else the selected tracks
`Ctrl+Alt+L` / `U`	Lock / unlock every track
Delete	Delete the selected clips (or tracks)

Key	What it does
`Opt+M` / `Opt+Shift+M`	Add a marker / a named marker
`Opt+R`	Region from the selection
`Shift+E`	Time selection from the selected clips
`[` / `]`	Previous / next marker or region
`Ctrl+Alt+M`	Show / hide the markers list
`Ctrl+Alt+R`	Sync from Console
`Ctrl+Alt+P`	Before You Record
`Cmd+N` / `Cmd+O` / `Cmd+S`	New / open / save
`Shift+Cmd+S`	Save As
`Cmd+I`	Import audio
`Cmd+E`	Export
`Cmd+Z` / `Shift+Cmd+Z`	Undo / redo
`Cmd+↑` / `Cmd+↓`	Move the selected tracks up / down
`Shift+Cmd+M`	Meters: show / hide
Number pad 0 / 3 / Enter	Play-stop / record / add a marker
Number pad `.` *n* `.`	Go to marker *n*
`Cmd,`	Settings
`Ctrl+Opt+K`	Keyboard Shortcuts (? until 0.2.6)

19. Acknowledgements

Show Recorder ships two libraries made by others, each as its own file in the app (`Contents/Frameworks/`), used through its public interface and unmodified:

- **LAME** by the LAME project — MP3 encoding, for Export and the mix during the take. LGPL-2.0 or later. https://lame.sourceforge.io/
- **libFLAC** by Xiph.Org — FLAC recording and playback. BSD-3-Clause; its licence is `COPYING.Xiph` in the app's `Contents/Resources/`. https://xiph.org/flac/

LAME's licence lets you replace the library with another build of LAME: swap the file in the Frameworks folder. Its source, as built into this app, is available on request from dylan@dylanmaudio.com.